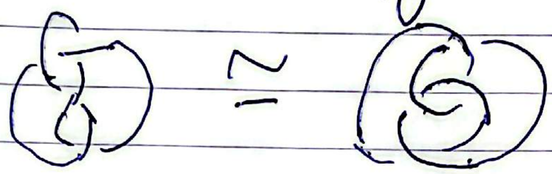
§1 Knot Invariants. ①

Defn A knot is a piecewise liner emb. $S^1 \to S^3$. w/ finitely many segments.

Theorems by people like Reidemeister show it's ok to replace "p.l." with eg. "smooth".

Natural eq. rel^n: ambient isotopy in S^3.

eg.  .

Qn. How to tell two knots apart? How to tell if a diagram represents the unknot?

Need to define invariants: computable functions

$i : \{\text{diagrams}\} \to \{\text{some nice set eg. } \mathbb{Z}, \text{polyn.}\}$

such that if $k \underset{\text{isot.}}{\simeq} l$, $i(k) = i(l)$.

Big industry eg. colourings, numbers, Alexander polynomial, homeo type*, hyp. volume, Jones poly, HOMFLY-PT, ...

Want invariants that are computable from a diagram.

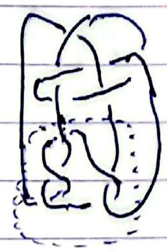
*Knots are comp. det. by homeo type. Gordon & Luecke, '89. (for knots). Not for links.

R. Riley '71: One family of invariants comes from reps

$$\rho : \pi_1(S^3 - k) \to PSL(2, \mathbb{F}_p)$$

and their related homologies.

e.g. reps into $PSL(2, \mathbb{F}_7)$ distinguish

Kinoshita-Terasaka '57  and  Conway '70.

which are indist. by Jones polynomials.

Qn. What about arbitrary reps

$$\rho : \pi_1(S^3 \setminus k) \to PSL(2, \mathbb{C})?$$

§2. $PSL(2,\mathbb{C})$.

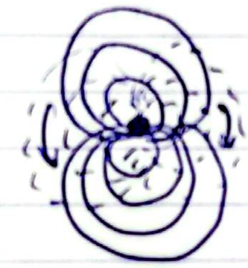
If $M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in PSL(2,\mathbb{C})$, not 1,

then M acts on $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\} = \mathbb{P}^1_{\mathbb{C}} = S^2$

by $M.z = \dfrac{\alpha z + \beta}{\gamma z + \delta}$.

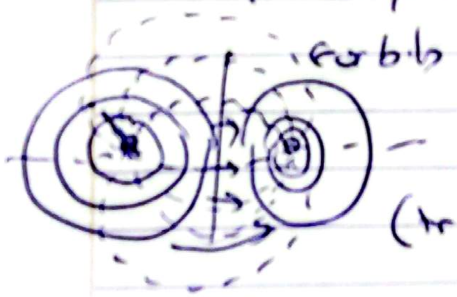
3 cases:

- $\mathrm{tr}^2 M = 4$ (parabolic):



Single fixed pt.

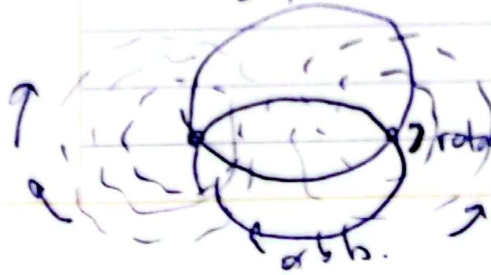
- $\mathrm{tr}^2 M \notin [0,4]$ (loxodromic):



($\mathrm{tr} \in \mathbb{R}$)

($\mathrm{tr} \notin \mathbb{R}$).

- $\mathrm{tr}^2 M \in [0,4)$ (elliptic):



rotate by angle θ.

($\mathrm{tr}\, M = 2\cos \theta/2$).

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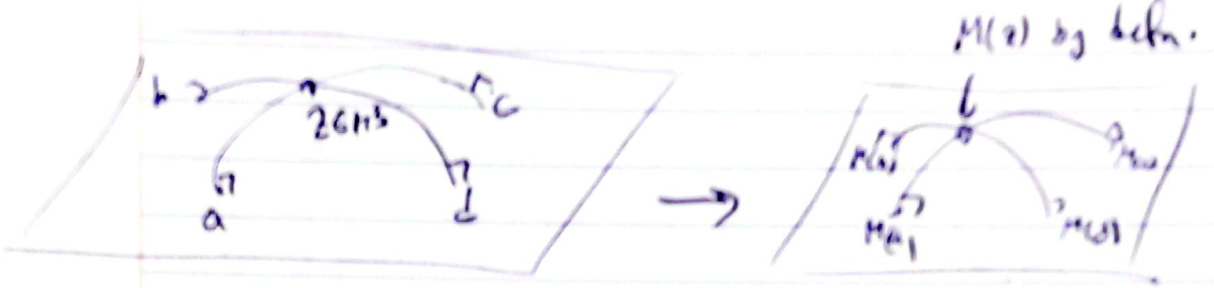
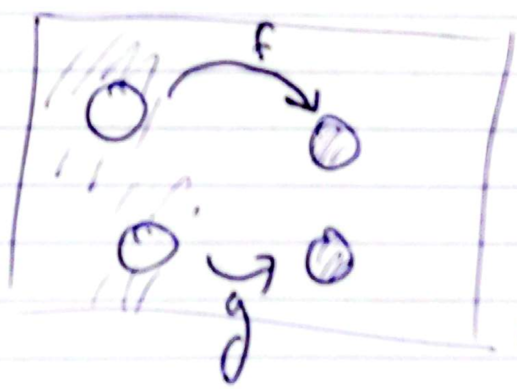
Poincaré observed that M also acts on $\mathbb{H}^3$:



This action is by isoms.

$\Rightarrow$ if $\Gamma < PSL(2,\mathbb{C})$ is discrete, then $\mathbb{H}^3/\Gamma$ is a hyperbolic orbifold

eg.



Thm. (Schottky, Fricke-Klein) if 2 pairs of disj. circles are given, and f, g send ext of one into the int. of the other in pairs, then $\langle f, g \rangle$ is discrete & free.

Also, $\mathbb{H}^3/\Gamma \cong$ .

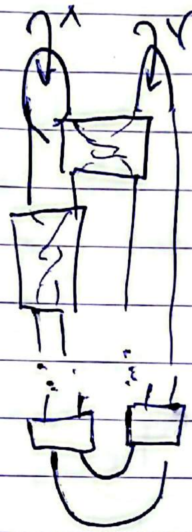
§3 The Riley slice.

(5)

Remember, Riley studied rep.

$$\pi_1(S^3 \setminus k) \to PSL(2,\mathbb{C}).$$

Consider ~~each~~ the 2-bridge link:



$$\pi_1(S^3 \setminus k) = \langle X, Y : W = 1 \rangle.$$

Thm (Riley): $\exists$ a discrete faithful rep. $\pi_1(S^3 \setminus k) \to PSL(2,\mathbb{C})$ iff k does not embed on a torus.

Cor. ('74): $\exists$ a knot k such that $S^3 \setminus k$ admits a complete hyp. metric

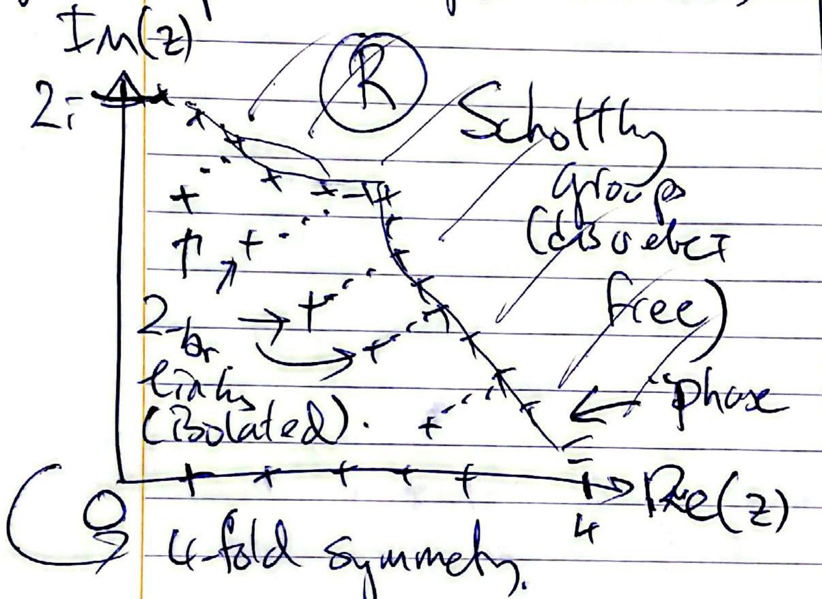
(major ~~most~~ influence on work of W. Thurston in late '70s).

~~Picture of all discrete reps of~~

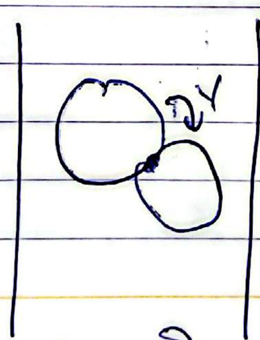
For hyp. geometry reasons, we consider only reps $\langle X, Y : \omega\rangle \to PSL(2,\mathbb{C})$ such that X, Y are both parabolic.

Riley's computer experiment ('71):

All subgps of $PSL(2,\mathbb{C})$ gen by 2 par. are $\left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix} \right\rangle$ up to conj.



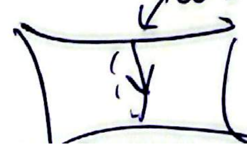
General group in the region R has fnd. dom. of the form



They are holonomies of hyperbolic str. on genus 2 handlebodies.

Thm (e.g. Ahlfors-Bers): R $= \mathrm{Teich}(S_{0,4})/\langle\omega\rangle$

ω = Dehn twist.



- Boundary of R is highly fractal & hard to understand.

- Giving bounds on R is very important, and has applications to:
  - enumeration of arithmetic groups in $PSL(2,\mathbb{C})$
  - computation of certain geometric minima (e.g. minimal volume manifolds).

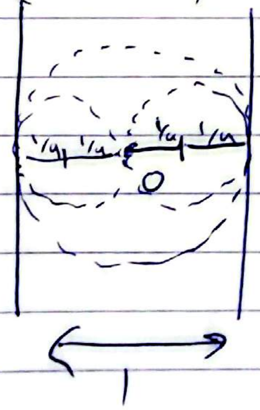
Basic bounds:

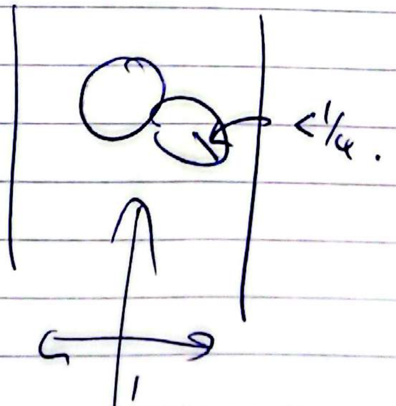
- Bremer ('55): When $|z| > 4$,

$$\Gamma_z = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix} \right\rangle$$ is discrete & free.

Proof: if $|z| > 4$ then the circles with centres $\pm 1/z$ and radii $1/|z|$ define a fundamental domain for Γ_z:

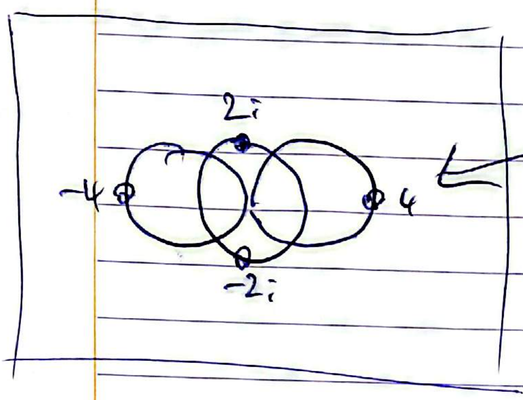
(extremal:)





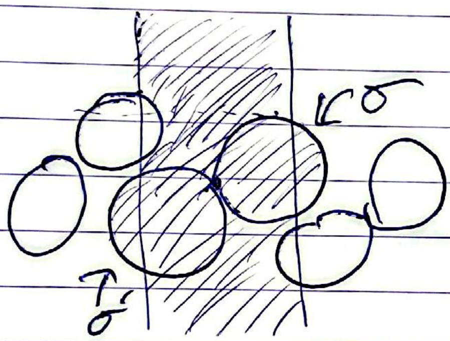
circles lie inside the strip.

Chang - Jening - Ree ('58):



$\leftarrow$ 3 inequalities from the condition that $X^{\pm 1}\left(B\left(\pm\frac{1}{2}, \frac{1}{|2|}\right)\right)$ are all disjoint:

Proof:

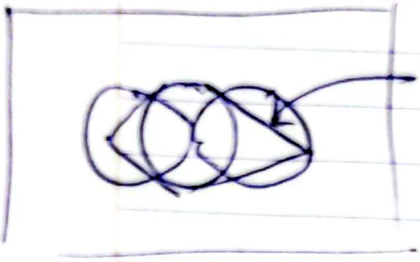


X sends shaded region into unshaded region.

Y sends □ into ▨ since it maps exterior of σ into interior of σ^{x}.

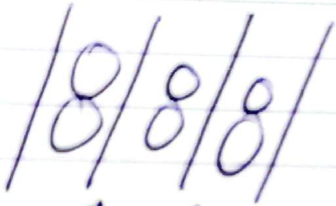
Ping-pong lemma: discrete & free.

- Lyndon-Ullman (1999):



improved to convex hull of $\{\pm 4, B(0,2)\}$.

Method: modification of ping-pong arg to



Shaded strips.

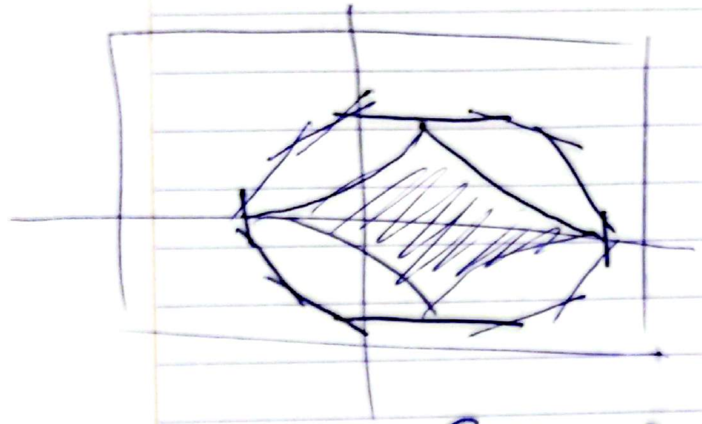
§4. Elliptic slices.

Defn. $R^{p,q} = \{ z : \Gamma_z = \langle \begin{pmatrix} e^{i\pi/p} & 1 \\ 0 & e^{-i\pi/p} \end{pmatrix}, \begin{pmatrix} e^{i\pi/q} & 0 \\ z & e^{-i\pi/q} \end{pmatrix} \rangle$ is discrete & iso. to $\mathbb{Z}/p\mathbb{Z} * \mathbb{Z}/q\mathbb{Z} \}$.

Every subgroup of $PSL(2,\mathbb{C})$ gen. by two elliptic matrices of order p, q is conjugate to some Γ_z.

Thm. (E.-Gong-Martin-Schillewaert, '24).
(to appear Ann. Inst. Fourier, '26).

$\mathbb{C} \setminus R^{p,q}$ is contained within a dodecagon:



A simple version. Fix $p=2$, $q=3$; this is motivated by studying discrete reps. of
$PSL(2,\mathbb{Z}) \simeq \mathbb{Z}/_{2\mathbb{Z}} * \mathbb{Z}/_{3\mathbb{Z}} \to PSL(2,\mathbb{C})$.

One choose of gens:

$\Gamma_\lambda := \left\langle u = \begin{pmatrix} 0 & \cancel{1} \\ \cancel{2\lambda}^{-1} & 0 \end{pmatrix},\ v = \begin{pmatrix} 1/2 & -\lambda\sqrt{3}/2 \\ \lambda^{-1}\sqrt{3}/2 & 1/2 \end{pmatrix} \right\rangle$.

Lem. Γ_λ is discrete & $\simeq \langle u \rangle * \langle v \rangle$
if the follow circles are disjoint:

f.d. for $\langle u, v \rangle$.

This is equiv. to

$$\left| \lambda \cot\frac{\pi}{3} \pm \cot\frac{\pi}{2} \right| + \csc\frac{\pi}{2} \le |\lambda| \csc\frac{\pi}{3}$$

i.e. $\frac{|\lambda|}{\sqrt{3}} + 1 \le \frac{2}{\sqrt{3}} |\lambda| \iff \sqrt{3} \le |\lambda|.$

$\cancel{|\lambda| + \sqrt{3} \le 2|\lambda|}.$

Another way of writing the generic rep^n of $PSL(2,\mathbb{Z}) \simeq \mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/3\mathbb{Z}$:

$$G_t := \left\langle \frac{1}{\sqrt{-t}} \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \frac{1}{\sqrt{-t}} \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix} \right\rangle.$$

This is essentially the image of the Burau rep^n of B_3 in $PSL(2,\mathbb{C})$.

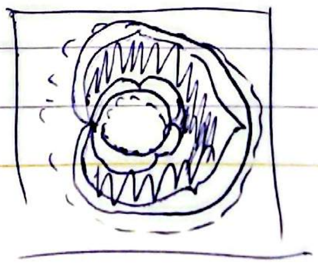
Change of var $t \leftrightarrow \lambda$:

$$\lambda = \frac{\sqrt{3}\left(\mu \pm \sqrt{\mu^2 + \mu + 1} - 1\right)}{3\sqrt{\mu}}.$$

Pushing $|\lambda| \ge \sqrt{3}$ through, we find that, if $z = \sqrt{\mu} - 1/\sqrt{\mu}$, and if

$$3 \le \left| z \pm \sqrt{z^2+3} \right|,$$

then $G_t \simeq B_3/Z(B_3)$.



Since $G_t = PSL(2,\mathbb{Z})_q$ for $q = -t$, we prove:

Conj. (Morier-Genoud, Ovsienko, Veselov): If $\frac{3-\sqrt{5}}{2} \le |q| \le \frac{3+\sqrt{5}}{2}$, then $PSL(2,\mathbb{Z})_q$ is fai. isom. to $SL(2,\mathbb{Z})$.